Technical Note

Power Electronics and Control Systems

Input-Series Output-Parallel

Phase-Shift Full Bridges

for a 3 kV Rail Auxiliary Supply

(model `ps_isop_rail`)

Averaged model, input-voltage sharing with a common phase shift,

why transformer pairs on one rectifier weaken it,

balancing controls, start and flux balance

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Abstract: The model `ps_isop_rail` is a 1 MW class auxiliary supply for a 3 kV DC catenary: four 250 kW phase-shift full bridges in series at the input and in parallel at the output (ISOP), with one output-voltage loop that gives the same phase shift to the four bridges and no input-voltage sharing loop. Its special feature is that the transformers work in pairs: the secondaries of two modules are paralleled on the AC side and feed one diode rectifier. This note derives the averaged model of a module, shows why a common phase shift balances the input voltages of modules on separate rectifiers (time constant 1.2 ms) and why the balance almost disappears between two modules on one rectifier: they are tied through leakage reactances, so a voltage difference drives a reactive circulating current and no restoring power. A switched simulation confirms it: 64 ms instead of 2.7 ms, and a 5 % leakage difference puts 1351 V and 1005 V on the two modules of a pair. Rev. 01 adds the controls that restore the balance: a loop per pair on the relative phase of its two bridges and a slow trim of the phase shift between the pairs bring the four voltages within 0.1 % in about 10 ms with 20 % more leakage in one module, while the classic per-module loop diverges with the modulator of the model. The note also covers the turns ratio, the start, the DC magnetizing current and the voltage loop. A companion Python script reproduces every number and figure.

Revision history

Rev.	Date	Description	Authors
00	November 2026	First issue.	Davide Bagnara
01	November 2026	Sect. 5: balancing controls (control diagram in Fig. 4, pair loops on the relative phase, pair-to-pair trim, why the classic per-module loop fails here); power in (4.2) with the factor 1/2 of peak phasors.	Davide Bagnara

Project Repository: modelization-and-control.git (foundations_of_power_electronics/ps_isop_rail)

Framework: MATLAB/Simulink/Simscape; Python 3 (companion script)

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1 Scope

The model `ps_isop_rail` (`foundations_of_power_electronics/ps_isop_rail` of modelization-and-control) is an auxiliary power supply for a 3 kV DC railway catenary. The catenary voltage (2000–3600 V of permanent range in EN 50163 [1]) is too high for one bridge with 1.7 kV devices, so four isolated modules share it in series at the input and give a common 750 V output in parallel (input series, output parallel: ISOP). Each module is a phase-shift full bridge (PSFB) [2]; one voltage loop gives the same phase shift to the four bridges (common-duty control [3]) and there is no input-voltage sharing loop.

During the migration of the model to the parameter structure of the repository (November 2026, pull request #26 of modelization-and-control) a design check with a switched simulation of the whole circuit was made. This note collects it: Sect. 2 gives the system, Sect. 3 the averaged model of a module and the turns ratio, Sect. 4 the input-voltage sharing (the main point), Sect. 5 the controls that keep it balanced (added in rev. 01), Sect. 6 the start, the flux balance and the voltage loop, Sect. 7 the summary. Every number comes from the companion script `isop_rail_auxiliary_supply_analysis.py`, which uses the switched simulator `isop_sim.py` (the same circuit and values as `init_model.m` of the model: backward Euler at 0.625 µs like the Simscape local solver, ideal switches and piecewise-linear diodes, the modulator and the voltage PI of the model).

2 The system

Fig. 1 shows the power stage and Table 1 the data. A lithium-ion battery emulates the catenary; it is set to 1.5 times the catenary voltage (4696 V open circuit), a choice that Sect. 3.2 explains. The four input capacitors are in series; each one feeds its own full bridge and transformer. The secondaries of modules 1 and 4 are paralleled on the AC side and feed rectifier 1, those of modules 2 and 3 feed rectifier 2; each rectifier has its own 250 µH inductor and the two share the output capacitor and the load.

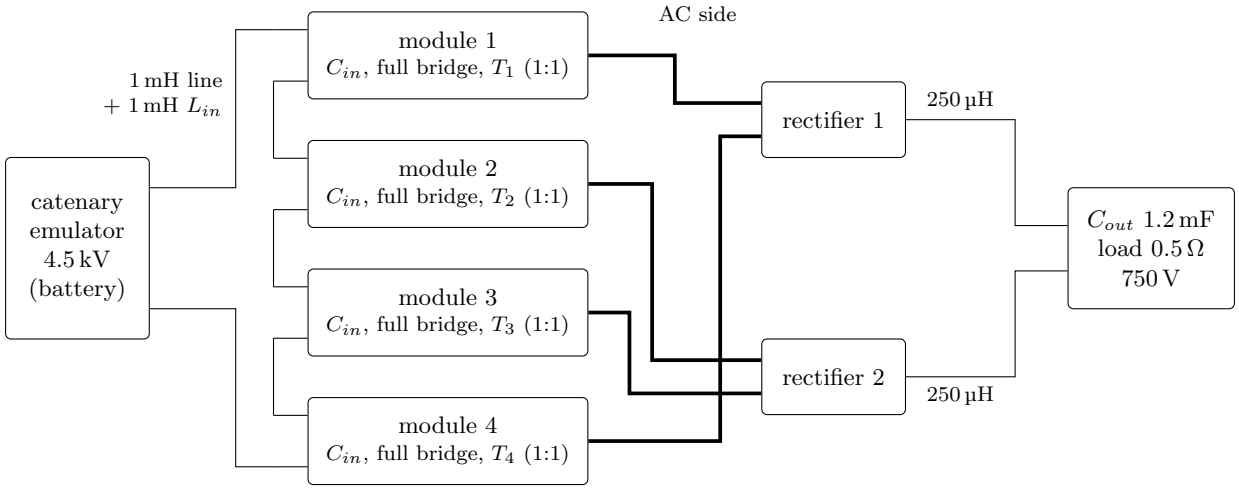


Figure 1: Power stage of `ps_isop_rail`: four modules in series at the input; the transformers of modules 1+4 and 2+3 are paralleled on the AC side (thick lines), one diode rectifier and one inductor per pair, common output capacitor.



Table 1: Data of the model (values of `init_model.m`).

Item	Value
catenary emulator	battery, 4500 V nominal, 4696 V open circuit (soc 0.85), 1 mH line; string inductor L_{in} 1 mH
module	250 kW; C_{in} 1.2 mF; full bridge of ideal switches (1.5 mΩ), 4 kHz, dead time 3 µs
transformer	$n = 1$, $L_m = 1$ mH, leakage $L_s = 26.0$ µH referred to the primary (half on each side), 1 mΩ per winding
output	two diode rectifiers, 250 µH each, C_{out} 1.2 mF, load 0.5 Ω: 750 V, 1500 A, 1.125 MW (112 % of 4×250 kW)
control	8 kHz (two updates per carrier period); voltage PI $k_p = 0.5$, $k_i = 18\,\mathrm{s}^{-1}$, output = phase shift $D \in [0, 1]$ (pu of π) of the four bridges; reference 750 V applied as a step

3 Averaged model of a module

3.1 Duty-cycle loss and output resistance

With the phase shift D the bridge puts $\pm v_k$ on the primary for the fraction D of each half period. The secondary current is the inductor current i_k (ripple neglected); at each change of polarity the primary current must go from $+n\,i_k$ to $-n\,i_k$ through the leakage, under the full voltage v_k, and during that time the rectifier short-circuits the secondary. The lost fraction of the half period is

$$\Delta D = \frac{2\,n\,i_k\,L_s}{v_k} \cdot 2f = \frac{4\,n\,L_s\,f\,i_k}{v_k}, \tag{3.1}$$

so the rectified output and the input current of module k are

$$v_o = n\,v_k\,(D - \Delta D) = n\,D\,v_k - R_{eq}\,i_k, \qquad R_{eq} = 4\,n^2\,L_s\,f, \qquad i_{in,k} = n\,(D - \Delta D)\,i_k = \frac{v_o\,i_k}{v_k}. \tag{3.2}$$

R_{eq} is a lossless resistance that models the duty-cycle loss [2]: here $R_{eq} = 0.417\,\Omega$, which costs 156 V at the 375 A of one module. At the operating point (1171 V per module) the phase shift is $D = 0.777$ (the switched simulation gives 0.782). Fig. 2 shows the switched waveforms there: the primary current reverses through the leakage at each change of polarity of the bridge voltage (the duty-cycle loss), the magnetizing current is small (52 A peak against 460 A), and the two secondaries of a pair carry the same current when the modules are equal. Each module has 1170.9 V and 380 A rms in its primary, and each rectifier gives 751.5 A.

3.2 Turns ratio, leakage and catenary voltage

With $n = 1$ and this leakage a module needs about 1170 V to give 750 V: that is why the emulator is set to 1.5 times the catenary. At the nominal 3 kV (750 V per module) the largest output at full load is 591 V; the simulation with the emulator at 3 kV stops at 604 V with the PI saturated (Fig. 6a). The leakage was sized with the formula of a dual active bridge (maximum power $V_1 V_2/(8 f L_s) = 2.7\,P$), but in a PSFB the leakage does not transfer power: it gives the soft switching and the duty-cycle loss. A PSFB is sized the other way round: the turns ratio from the lowest catenary voltage at which full output is needed, the leakage from an allowed duty-cycle loss. With a largest phase shift of 0.9 and 10 % duty-cycle loss: $n = 1.24$ and $L_s = 9.1$ µH for full output at 3 kV, $n = 1.85$ and $L_s = 4.05$ µH at 2 kV. The soft switching of the lagging leg at light load then sets a lower limit to L_s [2].

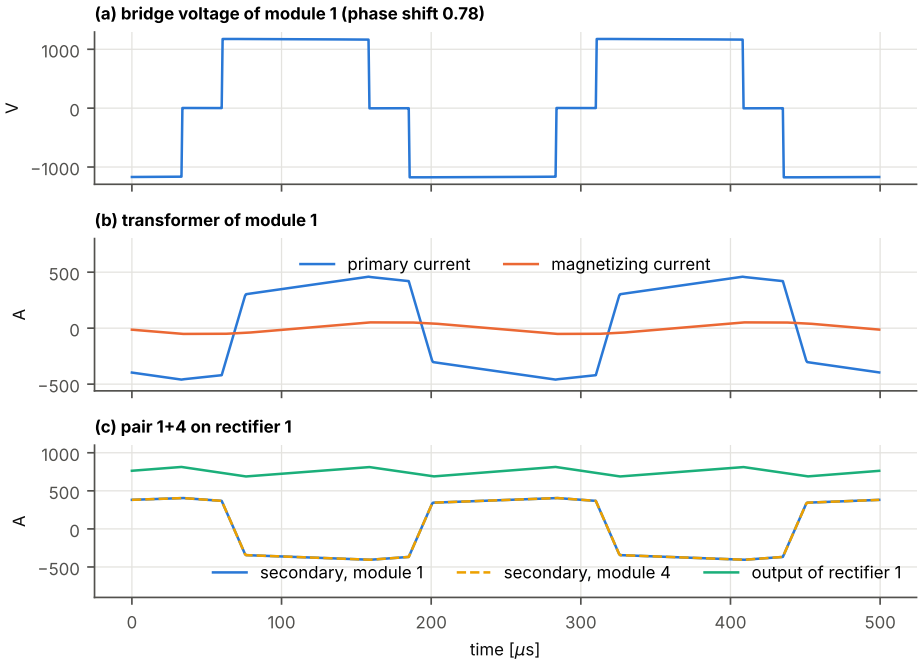


Figure 2: Steady state with identical modules (switched simulation): bridge voltage, primary and magnetizing current of module 1, secondary currents of the pair 1+4 and output current of rectifier 1.

4 Input-voltage sharing with a common phase shift

The modules are in series at the input, so they carry the same string current. In steady state each input capacitor has zero average current, so each module must draw the same average input current; the input voltages settle where this is true. Whether they settle at equal values, and how fast, depends on how a module's input current reacts to its own input voltage.

4.1 Modules on separate rectifiers: a resistive coupling

With separate rectifiers the modules meet on the DC side. With D common and v_o held by the output capacitor, (3.2) gives $i_k = (nDv_k - v_o)/R_{eq}$ and

$$\frac{\partial i_{in,k}}{\partial v_k} = g = \frac{v_o}{v_k}\left(\frac{n\,D}{R_{eq}} - \frac{i_k}{v_k}\right) = 0.6405\,(1.864 - 0.320)\,\mathrm{S} = 0.989\,\mathrm{S}. \tag{4.1}$$

A module whose voltage rises gives more output current, draws more input current and discharges its capacitor: every difference between the module voltages decays with $\tau = C_{in}/g = 1.2\,\mathrm{ms}$ (first order; the output inductors make the decay oscillating). This is the known self-balancing of ISOP modules with a common duty ratio [3]. The coupling goes through R_{eq}, which acts like a resistance: it moves real power from the module with the higher voltage. A mismatch only shifts the balance a little: in the averaged steady state a module with 20 % more leakage settles at 1208 V against 1158 V for the others (+3 %).

4.2 Two modules on one rectifier: a reactive coupling

In a pair the two transformers meet on the AC side, before the rectifier. With the fundamental-harmonic model each bridge is a voltage source in phase with the other (same carrier and phase shift), of peak value

$V_k = \frac{4}{\pi} v_k \sin(\pi D/2)$, connected through its leakage reactance $X_k = 2\pi f L_s n^2$ (referred to the secondary) to the rectifier input, of voltage $V_r\,\mathrm{e}^{-\mathrm{j}\theta}$ (peak phasors). The current and the real power of module k are

$$\underline{I}_k = \frac{nV_k - V_r\,\mathrm{e}^{-\mathrm{j}\theta}}{\mathrm{j}X_k}, \qquad P_k = \tfrac{1}{2}\mathrm{Re}\{nV_k\,\underline{I}_k^*\} = \frac{n\,V_k\,V_r\sin\theta}{2X_k}, \qquad i_{in,k} = \frac{P_k}{v_k} = \frac{2\,n\sin(\pi D/2)\,V_r\sin\theta}{\pi\,X_k}. \tag{4.2}$$

The input current does not depend on v_k: there is no restoring term. A difference $n(V_a - V_b)$ between the two bridges of a pair drives a circulating current $n(V_a - V_b)/(\mathrm{j}(X_a + X_b))$ in quadrature with the bridge voltages: reactive power, no real power. Only second-order effects (winding and switch resistances, the commutation of the rectifier) give some restoring power. And (4.2) shows that the leakage enters directly: the module with the larger X_k draws less input current, its capacitor charges and its voltage rises until the weak second-order terms balance the difference. A turns-ratio difference is harmless: the circulating current vanishes when $n_a v_a = n_b v_b$, and the pair settles there.

The four-winding transformer derived in `ps_isop_rail/trafo` (two primaries and two secondaries on one core) puts the two bridges of a pair on the same flux, so they are again tied through leakage reactances; it can save a core but it should have the same weak sharing (not simulated).

4.3 Switched simulation

Fig. 3 and Table 2 give the sharing; with identical modules it is perfect. With the phase shift fixed at the operating point and 80 V of initial deviation, the deviation between the two pairs decays with an envelope time constant of 2.7 ms (oscillating, through the output inductors), the deviation inside a pair with 64 ms, about 23 times slower. A 5 % leakage difference inside a pair puts 1351 V and 1005 V on its two modules (+15 % and −14 %); 20 % gives 1450 V and 959 V, and the module with the higher voltage carries about 357 kW (143 % of its rating) with 606 A rms and 900 A peak in its primary. The turns-ratio case settles at equal $n\,v$, as expected.

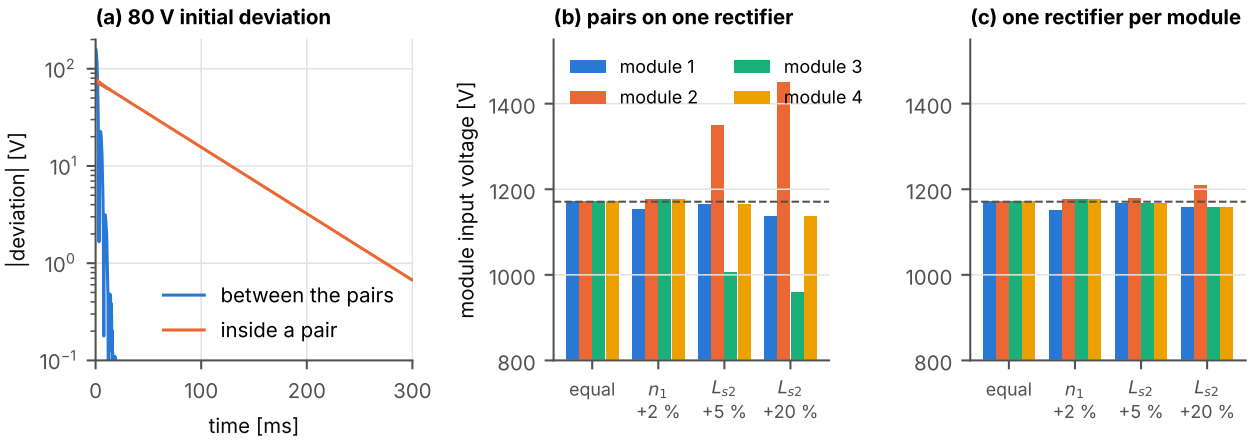


Figure 3: Input-voltage sharing. (a) Decay of an 80 V deviation at the operating point (phase shift fixed): between the pairs and inside a pair. (b) Steady-state module voltages of the model with one module different (switched simulation). (c) The same with one rectifier per module (averaged model (3.2)).

4.4 Remedies

1. **One rectifier per module.** Two more rectifiers and inductors, and the modules meet on the DC side again: the classic ISOP with the resistive coupling of the first subsection (+3 % for a 20 % leakage difference).

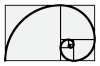


Table 2: Steady-state input voltages of the modules with one module different.

Case	Pairs on one rectifier (switched simulation)	One rectifier per module (averaged)
equal modules	1171 V on each module	1171 V on each module
module 1: n +2 %	module 1 1153 V, the others 1177 V	1150 V / 1178 V
module 2: L_s +5 %	module 2 1351 V, module 3 1005 V, 1 and 4 1164 V	1180 V / 1168 V
module 2: L_s +20 %	module 2 1450 V, module 3 959 V, 1 and 4 1137 V	1208 V / 1158 V

2. **Balancing loops** (Sect. 5). A duty correction changes mainly the amplitude of a bridge voltage, which in a pair gives reactive current again. A small time shift φ between the two bridges of a pair moves real power through the leakages, as in a dual active bridge [5]; a PI on $v_a - v_b$ that sets φ, plus a slow trim of the phase shift between the pairs, restores the sharing. The modulator then needs one carrier per bridge instead of one for all.

5 Controls that keep the input voltages balanced

Sect. 4 showed that a common phase shift keeps the two pairs balanced against each other, but hardly the two modules inside a pair. This section (added in rev. 01) proposes a control that does it, tests it in the switched simulation and explains why the classic per-module loop of the ISOP literature fails in this circuit. Fig. 4a shows the whole control, Fig. 4b what its two handles do in the modulator.

5.1 Two handles per bridge

Each bridge has two handles: its phase shift D, the width of its voltage pulses, which sets the amplitude V_k of (4.2); and the phase of its carrier (both legs moved together), which sets the angle of its voltage (Fig. 4b). Inside a pair the two bridges are two voltage sources on one AC node, each behind its leakage reactance, like two generators on one bus: real power follows the angle, while the amplitude moves mainly reactive power. If the voltages of the two bridges of a pair are moved by $+\varphi/2$ and $-\varphi/2$ in angle (carrier delays $\delta = \mp\varphi/2$), (4.2) gives

$$i_{in,a} - i_{in,b} \approx K_\varphi\,\varphi, \qquad K_\varphi = \frac{2\,n \sin(\pi D/2)\, V_r \cos\theta}{\pi X} = 838\,\mathrm{A/rad}, \tag{5.1}$$

with $V_r = 4v_o/\pi = 955\,\mathrm{V}$, $X = 0.654\,\Omega$ and the load angle $\theta = 16°$ (from P_k in (4.2)). The 44 A input-current difference that 20 % more leakage causes in one bridge (at equal module voltages) needs only $\varphi \approx 3°$, a time shift of 2 µs between the two carriers. A trim of the phase shift of one bridge, with its angle fixed, moves only $K_A = i_{in}\,(\pi/2)\cot(\pi D/2) = 138\,\mathrm{A}$ per unit: the same case would need ±0.16 on the two bridges.

5.2 Pair loops and pair-to-pair trim

The proposed control (Fig. 4a) adds three slow loops to the output-voltage loop, which keeps the common phase shift D:

$$\varphi_{14} = \mathrm{PI}_\varphi\{\bar v_1 - \bar v_4\}, \qquad \varphi_{23} = \mathrm{PI}_\varphi\{\bar v_2 - \bar v_3\}, \qquad d = \mathrm{PI}_d\{(\bar v_1 + \bar v_4) - (\bar v_2 + \bar v_3)\}. \tag{5.2}$$

The carrier of bridge 1 moves earlier by $\varphi_{14}/2$ and that of bridge 4 later by $\varphi_{14}/2$ (the same for bridges 2 and 3 with φ_{23}), so the bridge with the higher input voltage leads and takes more power; pair 1+4 gets the phase shift $D + d$ and pair 2+3 gets $D - d$. Here $\bar v_k$ is the input voltage of module k averaged over one switching period (the mean of the last two samples, see below). The pair loop sees the plant $C_{in}\,\mathrm{d}(v_a - v_b)/\mathrm{d}t = -K_\varphi\,\varphi$: with $k_p = 3.6 \times 10^{-4}\,\mathrm{rad/V}$ and $k_i = 0.0216\,\mathrm{rad/(V\,s)}$ it crosses over at 250 rad/s (40 Hz), its zero is at 60 rad/s,

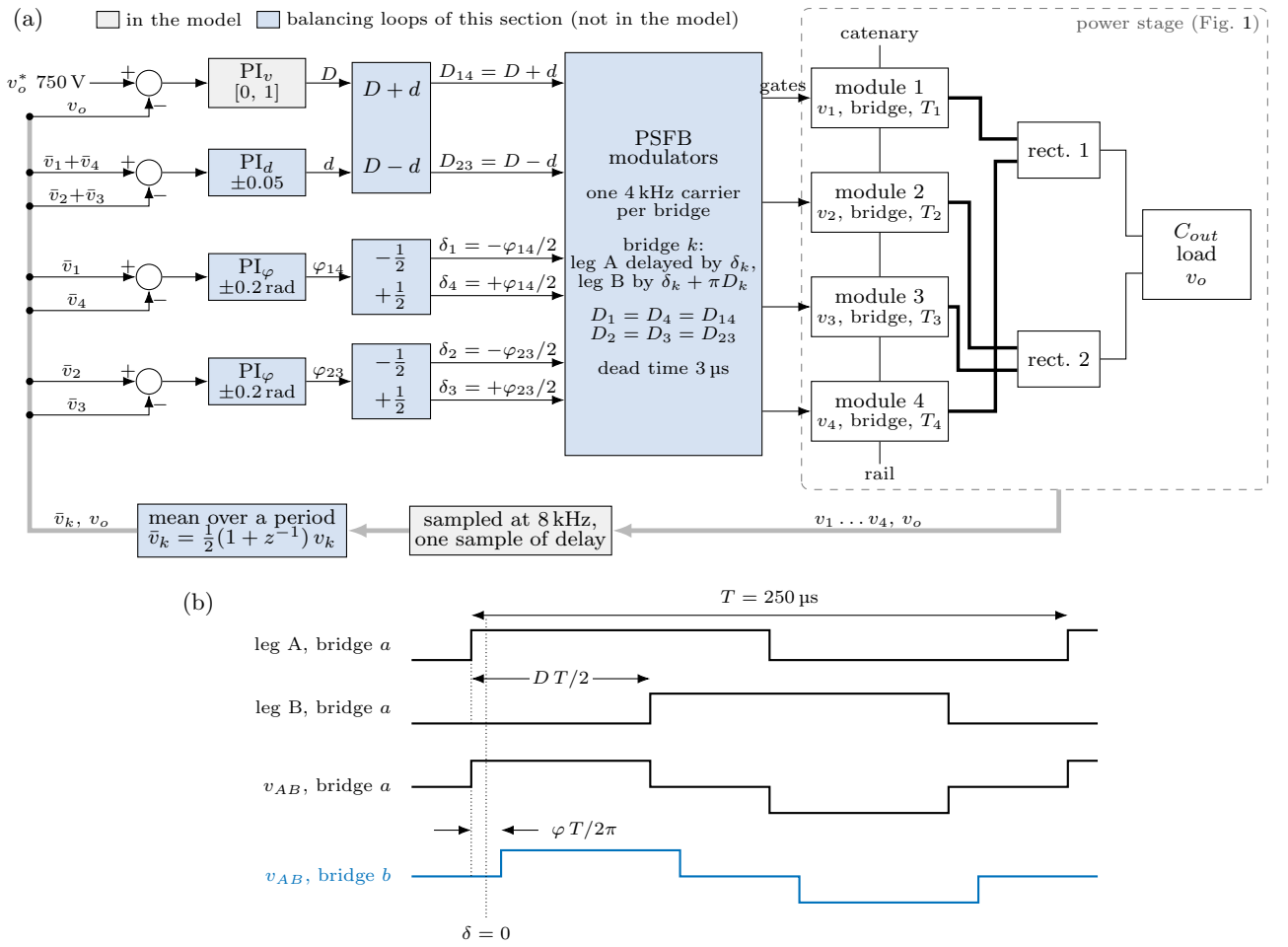


Figure 4: Control with the balancing loops. (a) Block diagram: the output-voltage loop of the model gives the common phase shift D; the balancing loops of this section add a trim $\pm d$ of the phase shift between the pairs and, inside each pair, opposite carrier delays $\delta_k = \mp\varphi/2$ (rad of the switching period, negative: earlier) from a PI on the voltage difference of its two modules. All loops run at 8 kHz with one sample of delay; the module voltages are averaged over one switching period. Thick lines: AC side. (b) Modulation of the two bridges a and b of a pair (sketch with $D = 0.6$, shift exaggerated): the phase shift moves leg B only and sets the width $D\,T/2$ of the voltage pulses (the amplitude); the carrier delays $\delta_a = -\varphi/2$ and $\delta_b = +\varphi/2$ move both legs and shift the pulses by $\varphi\,T/2\pi$ (the angle).

and φ is limited to $\pm 11.5°$. Between the pairs the resistive coupling of Sect. 4 already acts, so the trim d only removes the static error left by a mismatch, as in the common-duty control with active sharing of [3]: from (3.2) each module draws $n v_o / R_{eq} = 1800$ A more per unit of D, and with $k_p = 1 \times 10^{-4}\,\mathrm{V}^{-1}$, $k_i = 0.014\,\mathrm{V}^{-1}\,\mathrm{s}^{-1}$ and the limit ± 0.05 the closed-loop poles are at -62 rad/s and -1362 rad/s. The phase offsets are opposite inside a pair and the trims are opposite between the pairs, so the output-voltage loop does not see them.

5.3 Why the classic per-module loop fails here

The classic active input-voltage sharing loop [3, 4] trims the phase shift of each module from v_k minus the mean module voltage. Inside a pair this has two problems. The trim acts on the amplitude, the weak handle (K_A above). And the modulator of the model keeps leg A fixed and delays leg B by $D\,T/2$ (Fig. 4b), so a trim x also delays the centre of the voltage pulse by $x\,T/4$, an angle of $\pi x/2$. For trims $\pm x$ on the two bridges of a pair the angle wins and the sign is wrong: $\partial(i_{in,a} - i_{in,b})/\partial x = -\pi K_\varphi + 2K_A = -2359$ A. The module with the

higher voltage gets a longer pulse, lags its partner and takes even less power. The switched simulation confirms it (Fig. 5c): 20 ms after the loop starts, module 3 is down to 8 V and module 2 is up to 2.33 kV, with 2.7 kA peak in its primary. With a symmetric modulator (legs moved by $\mp D\,T/4$, pulse centre fixed) the sign is right, but the action is 9.6 times smaller than the phase effect: with trims limited to ±0.1 the deviation falls only from 24 % to 7.3 % (Fig. 5d). Between modules on separate rectifiers the classic loop works, because there the angle does not matter.

5.4 Switched simulation

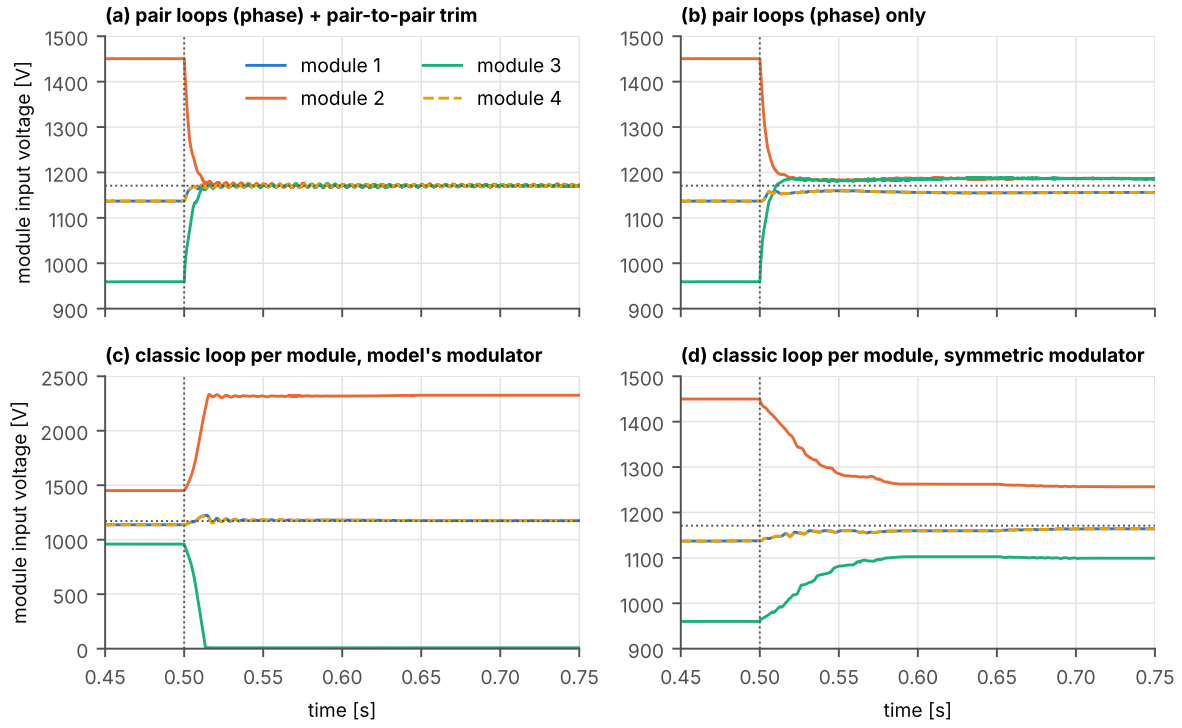


Figure 5: Module input voltages with 20 % more leakage in module 2; the balancing loops start at 0.5 s (switched simulation). (a) Pair loops on the relative phase and pair-to-pair trim of the phase shift. (b) Pair loops only. (c) Classic per-module loop on the phase shift with the modulator of the model (leg B moved): it diverges. (d) The same loop with a symmetric modulator: right direction, too weak.

Table 3: Balancing controls in the switched simulation: module input voltages 0.5 s after the loops start, largest deviation from the mean, relative phase of the pair and phase-shift trim, time to come within 2 %.

Case	Control	v_1, v_2, v_3, v_4 [V]	Deviation	φ, d	Within 2 %
L_{s2} +20 %	none	1137, 1450, 959, 1137	23.9 %	–	–
	pair loops + trim	1171, 1171, 1171, 1171	0.07 %	2.5°, ±0.007	10 ms
	pair loops only	1157, 1186, 1185, 1157	1.2 %	2.5°, –	16 ms
	classic, model modulator	1173, 2328, 8, 1173	99 %	–, at the limit	diverges
	classic, symmetric modulator	1170, 1256, 1088, 1170	7.3 %	–, at the limit	never
L_{s2} +5 %	pair loops + trim	1172, 1172, 1171, 1172	0.16 %	0.7°, ±0.004	5 ms
n_1 +2 %	pair loops + trim	1170, 1170, 1170, 1171	0.17 %	0°, ±0.003	(1.5 % before)
equal modules	pair loops + trim	1171 on each module	0	0, 0	–

The loops start at 0.5 s, from the unbalanced steady state of Sect. 4 (Fig. 5, Table 3). With 20 % more leakage in module 2 the four input voltages come within 2 % in 10 ms and end within 0.07 %, with $\varphi_{23} = 2.5°$, against

3.0° from (5.1), and $d = \pm 0.007$. The primary currents fall from 606 A rms and 900 A peak in module 2 to 369–389 A rms and at most 514 A peak. The pair loops alone balance each pair but leave 1.2 % between the pairs. With identical modules the loops do nothing.

5.5 Practical notes

- **Average the measured voltages over a switching period.** The control samples twice per period. A trim that differs between the two half periods gives unequal volt-seconds and a DC magnetizing current; the DC current makes the two half periods of the input current different, and the next samples see that difference. Without the average, the case with 2 % more turns in module 1 builds up −115 A of DC magnetizing current in modules 1 and 4 in 0.5 s, still growing; with the average it stays at the few amperes left from the start. The phase offsets move both legs together and do not change the volt-seconds; the trim of the phase shift does.
- **Modulator.** One carrier per bridge with a phase input, and a phase-shift input per pair. The offsets are small (2.5° is 1.7 µs at 4 kHz), so the PWM needs a fine time resolution. A DSP timer has it; the simulation with its 0.625 µs step still works because the loop dithers between two steps.
- **Turns mismatch.** The pair loop makes the input voltages equal, not the secondary voltages: with 2 % more turns in module 1 a reactive current circulates in the pair (module 1: 393 A rms and 501 A peak instead of 381 A and 463 A). Within normal tolerances this is small.
- With one rectifier per module no pair loop is needed: the classic per-module loop, or no loop at all, is enough.

6 Start, flux balance and voltage loop

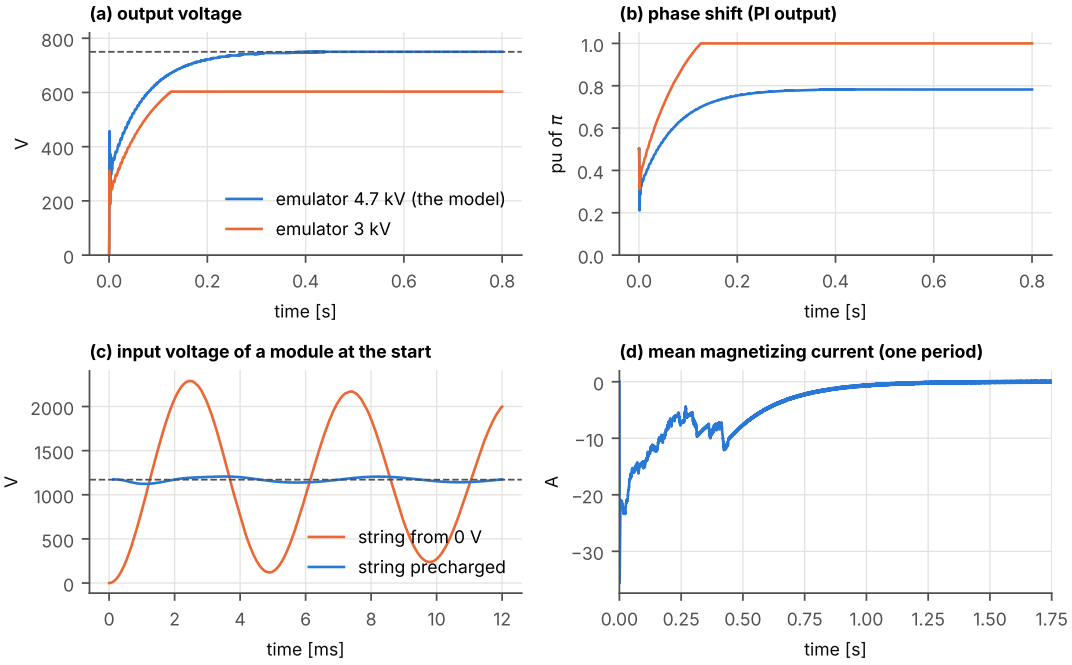


Figure 6: Start (switched simulation). (a) Output voltage and (b) phase shift with the emulator of the model and with the emulator at 3 kV. (c) Input voltage of a module with the string precharged and from 0 V. (d) Mean magnetizing current of a transformer.

Start. The model starts from the DC steady state of the network (string precharged); the output reaches 95 % of 750 V in 178 ms without overshoot (Fig. 6a,b), and the input filter (2 mH with the four capacitors in series, 205 Hz) rings a little at the start. If the string capacitors start at 0 V (their initial target in the model), the input filter rings up to 2290 V per module, twice the nominal, with 1.8 kA in the string (Fig. 6c): the real system needs a precharge of the string and a soft start of the reference.

DC magnetizing current. The phase shift changes at every half period during the start; each change between a positive and a negative half period gives unequal volt-seconds on the transformer and a step of DC magnetizing current. They add up to −21 A, 40 % of the 52 A AC peak, which decays only through the winding and switch resistances ($L_m/R \approx 0.2$ s, Fig. 6d); with a real core this is a saturation risk. The "DC transformer current" PI of the model cannot correct it: its output is not connected, it measures the input current of the string (a DC current by nature) and it acts on the duty of both legs together, which gives no DC voltage on the transformer. A flux balance needs the DC part of each primary current and a differential action on the two legs; the usual alternatives are a DC-blocking capacitor in series with each primary or peak-current-mode control.

Voltage loop. With the averaged plant (gain $n\,v_k/750$ V reduced by the droop of R_{eq}: 1.29 pu/pu), the output filter and 1.5 samples of delay, the gains of the model give a crossover of 4.8 Hz (phase margin 129°, gain margin 9.1 dB, closed-loop pole −14 rad/s). The proportional gain is limited by the resonance of the output filter at 411 Hz: with $k_p = 0.8$ the loop crosses over at the resonance (511 Hz, gain margin 4.9 dB). A faster loop needs more integral gain: $k_p = 0.6$, $k_i = 100\,\mathrm{s}^{-1}$ gives 33 Hz, 135°, 7.2 dB and a pole five times faster. Faster still, the input filter (characteristic impedance 2.6 Ω) must be checked against the −19.5 Ω constant-power input of the converter [6].

7 Summary

- A phase-shift module behaves on average like nDv_k behind the lossless resistance $R_{eq} = 4n^2L_sf$; with a common phase shift this resistance balances the input voltages of ISOP modules on separate rectifiers in about a millisecond.
- Two modules whose transformers are paralleled on the AC side are tied through leakage reactances, so a voltage difference gives only reactive current: in `ps_isop_rail` the sharing inside a pair is about 23 times slower and a 5 % leakage difference splits the voltages by ±15 %. One rectifier per module or a pair loop on the relative phase fixes it; a common-core four-winding transformer does not.
- A loop per pair on the relative phase of its two bridges (a few degrees, 40 Hz crossover) plus a slow trim of the phase shift between the pairs keeps the four input voltages within 0.2 %, even with 20 % more leakage in one module. The classic per-module loop on the phase shift has the wrong sign inside a pair with the modulator of the model. The measured voltages must be averaged over a switching period, or the trims build up a DC magnetizing current.
- With $n = 1$ and a DAB-style leakage the output reaches 750 V only with the emulator at 1.5 times the catenary; a PSFB needs the turns ratio from the lowest catenary voltage and a small leakage.
- The start needs a precharge; the DC magnetizing current left by the start is not controlled; the voltage loop is slow and its proportional gain is limited by the output filter.

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